(Please write your Enrollment Number) Enrollment No. 11001172022

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech>, Semester:<5>
(December, 2024)

Subject Code: BCS 303	Subject: Theory of Computation
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)	CO Mapping
	a) Why $a^* b^*$ is regular but not $a^n b^n$?		CO 1
	b) Define Deterministic Finite Automata (DFA)? How is it different from Non Deterministic Finite Automata (NDFA).		CO 1
	c) What is Derivation Tree and its types? Discuss with an example.		CO 2
	d) Define Context Free Grammar. Create a CFG for $\{a^n b^n : n \geq 0\}$		CO 3
	e) Define the term ambiguity and explain with the help of an example grammar?		CO 1
	f) Define Unrestricted Grammar (UG). Write about the machine that accepts the UG.		CO 2
	g) What is halting problem, discuss about it briefly?		CO 3
	h) With neat diagram discuss about Linear Bounded Automata.		CO 1
	UNIT I		**CO Mapping**
Q2	Design a DFA for the Language accepting strings starting with 'aba' and ending with 'bb' over the alphabet Σ = {a, b}. Check whether the DFA accepts the strings: abaabb, abbbaab, ababa.	(10)	CO 2 & CO 3
Q3	Discuss the step by step procedure followed in minimization of Deterministic Finite Automata and minimize the following finite automata.	(10)	CO 2

	0	1
$\rightarrow q_0$	q_1	q_4
q_1	q_5	q_2
$*q_2$	q_0	q_2
q_3	q_6	q_4
q_4	q_2	q_5
q_5	q_5	q_3
q_6	q_5	q_2

Find out whether the minimized DFA accepts the following strings:
0010, 1101, 10010.

	UNIT II		**CO Mapping**
Q4	Define a Pushdown Automata (PDA) (describe each tuple). Design a PDA which accepts the Language $L = \{0^n 1^n / n > 1\}$ over $\Sigma = \{0, 1\}$.	(10)	CO 1 & CO 2
Q5	What is Chomsky Normal Form (CNF)? Convert the following Context Free Grammar to CNF? (where ε is epsilon) S→ABa / BBb / DB / DA A→aab / bba / a B→Ac / C /b C→c / d / F / H D→d / ε	(10)	CO 3

<table>
<tr><td colspan="2">UNIT III</td><td></td><td>CO Mapping</td></tr>
<tr><td>Q6</td><td>Define pumping Lemma for Context Free Language. And check whether the language L = { $0^n1^n2^n3^n$/ n>1} is context free or not.</td><td>(10)</td><td>CO 1 & CO 2</td></tr>
<tr><td>Q7</td><td>Design a TM for the Language L = {$a^nb^nc^n$ / n>1}.</td><td>(10)</td><td>CO 1 & CO 3</td></tr>
<tr><td colspan="2">UNIT IV</td><td>CO Mapping</td><td></td></tr>
<tr><td>Q8</td><td>What is Finite state Transducer? Explain. the step by step procedure followed for converting a Mealy Machine to Moore Machine. Convert the following Mealy Machine to Moore Machine.
<table>
<tr><th rowspan="2">Present state</th><th colspan="4">Next state</th></tr>
<tr><th colspan="2">Input a=0</th><th colspan="2">Input a=1</th></tr>
<tr><th>State</th><th>output</th><th>State</th><th>output</th></tr>
<tr><td>→q_1</td><td>q_3</td><td>0</td><td>q_2</td><td>1</td></tr>
<tr><td>q_2</td><td>q_1</td><td>0</td><td>q_4</td><td>0</td></tr>
<tr><td>q_3</td><td>q_2</td><td>1</td><td>q_1</td><td>1</td></tr>
<tr><td>q_4</td><td>q_4</td><td>1</td><td>q_3</td><td>0</td></tr>
</table></td><td>(10)</td><td>CO 1 & CO 3</td></tr>
<tr><td>Q9</td><td>Discuss about Complexity classes P and NP. Define a Moore Machine (using tuples). Convert the following Moore Machine to Mealy Machine.
<table>
<tr><th rowspan="2">Present State</th><th colspan="2">Next state</th><th rowspan="2">Output</th></tr>
<tr><th>a=0</th><th>a=1</th></tr>
<tr><td>→q_0</td><td>q_3</td><td>q_1</td><td>0</td></tr>
<tr><td>q_1</td><td>q_1</td><td>q_2</td><td>1</td></tr>
<tr><td>q_2</td><td>q_2</td><td>q_3</td><td>0</td></tr>
<tr><td>q_3</td><td>q_3</td><td>q_0</td><td>0</td></tr>
<tr><td>q_4</td><td>q_0</td><td>q_4</td><td>1</td></tr>
</table></td><td>(10)</td><td>CO 1 & CO 3</td></tr>
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